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Bihar Engineering University, Patna

B.Tech 2nd Semester Examination, 2026

Time: 03 Hours

Code: 100204 ***Subject: Elements of Mechanical Engineering*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option/answer the following (Any seven question only): **[2×7=14]**

(a) Which of the following is not a renewable source of energy?
(i) solar (iii) fossil fuels
(ii) hydropower (iv) wind

(b) Which of the following represents an intensive property
(i) volume (iii) mass
(ii) temperature (iv) energy

(c) Which one is the accessory of the boiler?
(i) steam injector (iii) fusible plug
(ii) pressure gauge (iv) blow of cock

(d) Which is not a part of vapour compression refrigeration system?
(i) compressor (iii) absorber
(ii) receiver (iv) Throttle valve

(e) Air-conditioning means the control of
(i) temperature (iii) purity of air
(ii) humidity (iv) all of these

(f) In C.I. engine working on diesel cycle, during the suction stroke
(i) only fuel is sucked (iii) only air is sucked
(ii) mixture of fuel and air is sucked (iv) none of these

(g) The Otto cycle is also known as
(i) constant volume cycle (iii) constant pressure cycle
(ii) constant temperature (iv) constant enthalpy cycle

(h) Which of the following energy is converted into electricity in a hydro power plant?
(i) Nuclear energy (iii) Thermal energy
(ii) potential energy of water (iv) All of these

(i) 'Tempering' of quenched martensitic steel is necessary to improve the
(i) Hardness of the metal (iii) Ductility of the metal
(ii) Corrosion resistance of the metal (iv) Surface texture of the metal

(j) The property of a material which enables it to resist penetration or abrasion is called
(i) toughness (iii) stiffness
(ii) corrosion resistance (iv) hardness

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Q.2 (a) Distinguish between non-renewable and renewable energy resources with suitable example. **[7]**

(b) What is biomass? How the biomass energy is obtained? **[7]**

Q.3 (a) Define thermodynamic system and its properties, state and processes. What do you mean by a thermodynamic cycle? **[7]**

(b) Air initially at a pressure of 75 kPa and temperature of 1000 K and occupying a volume of 0.12 m^3 is compressed isothermally until the volume is halved. Subsequently, the volume undergoes further compression at constant pressure till it is halved again. Calculate the work done. **[7]**

Q.4 (a) Differentiate between internally and externally fired boilers. **[4]**

(b) Explain the difference between boiler mounting and accessories. **[4]**

(c) Give the compression between fire tube and water tube boiler. Can a fire tube boiler be used for high pressure steam generation? Explain with reasons. **[6]**

Q.5 (a) Describe with a neat diagram the essential components of vapour compression refrigeration system. **[7]**

(b) Sketch and explain a generalized air conditioning system. **[7]**

Q.6 (a) Explain the working operation of a closed cycle gas turbine with neat sketches. Show all the processes on p-V and T-S diagram. **[7]**

(b) Differentiate between two stroke and four-stroke engines. **[7]**

Q.7 (a) Explain with a neat sketch the working principle of nuclear power plant. **[7]**

(b) Explain with a neat sketch the working principle of thermal power plant. **[7]**

Q.8 (a) Distinguish between ferrous and non-ferrous metals. **[8]**

(b) Define the following properties of material (i) Hardness (ii) Strength (iii) Toughness **[6]**

Q.9 (a) What are the advantages of heat treatments? **[7]**

(b) Define and explain the heat treatment process, annealing, tempering and quenching? **[7]**

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